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AI - CO1 - AT-2

① HOSPITAL DOCTOR SCHEDULING USING BACKTRACKING SEARCH :-

• CONSTRAINTS :-

→ D1 can't work the NIGHT shift.

→ D_2 must work before D_3.

→ Only one doctor can be assigned to each shift

→ D_3 can't work the morning shift.

• Step-by-step-Solution :-

① Assign D_2 to Morn :- This satisfies the requirements that D_2 works before D_3.

② Assign D_1 to Aftn :- D_1 is not assigned to the Night Shift, so this constraint is satisfied.

③ Assign D_3 to Night :- D_3 doesn't work in the Morning & D_2 works before D_3.

④ Check constraints :- All three docs have different shifts & every doctor is assigned exactly once.

• Backtracking Explanation :-

→ It assigns a value to a variable & checks whether the constraints are satisfied. If an assignment violates a constraint, it is undone & another value is tried.

(2) Robot Navigation Using BFS SEARCH ALGORITHM :-

• Constraints :-

→ Movement is alloweded UP, Down, Left & Right

→ Each move has a cost of 1.

→ Obstacles can't be crossed.

→ Manhattan distance is used as the heuristic.

• Step-by-step soln :-

① Initialize :- Place the start node S in the queue & mark it as visited.

② Expand nodes :- Remove the front node from the queue & explore its valid neighbouring cells in the order up, down, left & Right.

③ check constraints :- Reject cells that are outside the grid, contain obstacles, or have already visited.

④ Repeat :- Continue expanding nodes until goal is reached.

⑤ Find the path :- Trace the parent nodes backwards from G to S to obtain the shortest path.

→ $h(n) = |x_n - x_{G_1}| + |y_n - y_{G_1}|$

• Optimal path & cost :- BFS finds the shortest path in terms of the number of moves when every move has equal cost. The Total Path cost is

• Path Cost = No. of Moves × 1.

→ The exact node expansion sequence, queue, updates & final path can't be determined because the question paper doesn't show the positions of the obstacles S or G.

(3) Autonomous Rescue Robot in a Disaster-Affected Building :-

(a) Analyze the constraints :-

→ The robot must reach the survivor at G from S while minimizing cost & ensuring safety.

• Movement :- only alloweded up, down, Left & Right.

• Obstacles :- Blocked cells must be avoided.

• Limited Sensing :- The robot can only observe adjacent cells.

• Dynamic knowledge :- The robot must update its map as it discovers new information.

(b) Soln using UCS & Online Search agent :-

(1) Start at S with cost of zero.

(2) Observe adjacent cells & identify obstacles & risky zones.

(3) Add valid neighbouring cells with accumulated costs.

④ Select cell with the lower total cost using UCS.

⑤ Update the map & replan whenever new info is discovered.

⑥ Repeat until Robot reaches G through minimum cost

© Application of AI concepts :-

- Intelligent Agent :- The robot senses its surrounding & selects actions to reach the survivor.
- Rationality :- It chooses action that minimizes cost
- Type :- The environment is partially observable, sequential, discrete & initially unkown to robot.

ⓓ Justification :-

→ UCS is suitable because it considers the total path cost, including additional cost for risky zones, & finds a minimum-cost path when movement costs are non-negative. The online search agent helps the robot discover obstacles, update its knowledge & adjust its route.